

## **Infrastructure planning**

101-0509-10L (ECTS-6KP)

## **Appraising infrastructure modifications and adaptive plans**

### **Project Sheet 1 (Weeks 1 and 2)**

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## 1 Background

Infrastructure modifications are often required to ensure that infrastructure continues to meet the changing needs of society. For example, when there are more people who want to travel by bicycle or e-bike than before and/or society wants to encourage people to travel by bicycle or e-bike, it is reasonable to consider the expansion of the cycling network to reach more areas. Before infrastructure modifications are undertaken, possible alternatives need to be appraised so that stakeholders can decide which of the modifications they prefer most, taking into consideration how the infrastructure system may change in the future with and without the modification.

Appraising potential infrastructure modifications and associated adaptive plans (i.e., plans that acknowledge the ability to take different actions in the future depending on the actual situation) can be done by estimating the probabilities of future scenarios occurring and valuing costs and benefits. When it is not possible to estimate the probabilities of future scenarios or value costs and benefits, it is possible to appraise modifications using subjective valuations, explicit trade-offs between objectives and measures of robustness (i.e. how well the plan performs under many potential scenarios). In reality, appraisals are often a combination of the two approaches.

## 2 Project Overview

While appraising infrastructure modifications in the real-world takes considerable amounts of time and would not be possible to replicate in a course environment, we have devised two projects that will enable you to gain an appreciation of how this is done at a moderately advanced level. The first project is the appraisal of the Limmattal bike highway (Veloschnellroute Limmattal)<sup>1</sup>. The second project is the appraisal of the Küsnacht - Bahnhof Forch Mobility Hub (Verkehrsdrehscheibe)<sup>2</sup>.

For both, we are assuming you are in the strategic planning part of the SIA102 process and a preliminary appraisal of the potential infrastructure modifications and adaptive plans has already been made. These preliminary appraisals (based on previous year's projects) will be provided as a starting point for understanding the case study, the analyses previously conducted, and the arguments used to support planning decisions. Although you find the preliminary appraisal valuable and insightful, you realise that a more in-depth analysis must be done before a decision can be made. Therefore, you are expected to critically assess and improve them, making use of the more advanced tools and methods explored in this year's course. In particular, you should consider how these tools can strengthen both the analysis itself and the argumentation underlying how an infrastructure planning problem should be approached and how alternative infrastructure modifications and adaptive plans should be evaluated.

To conduct the more in-depth analysis, you will work in groups of 3. The improved analysis you conduct will take into consideration our suggestions, but will also include some of your own; i.e., you will expand the analysis in a way that you are convinced you will be able to determine which infrastructure modifications and adaptive plans to propose. Your final argument will present the system changes you expect for a range of believable future scenarios, including the respective outcomes, costs and benefits. It will also consider qualitative aspects that cannot be fully captured by the quantitative analysis, in order to provide a more comprehensive basis for the final recommendation. The final report should be less than 30 pages (main body without references).

By doing one of these two projects and participating in the discussion at the end of the year, the entire class will obtain familiarity with transport models, uncertainty analysis and appraisal methods. To help your peers understand your improvements to the preliminary appraisal, you will be expected to succinctly explain one or more of them and the effect they had on the results and the decision. By presenting these ideas to the class you will gain experience in presenting your ideas crisply and cleanly to peers, and through giving constructive feedback, you will also gain such experience.

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<sup>1</sup> [Velonetzplanung | Kanton Zürich](#)

<sup>2</sup> [Küsnacht – Mobilitätsdrehscheibe Bahnhof Forch | Kanton Zürich](#)

## 2.1 Specifics for the Bike Highway Limmattal Project

### 2.1.1 Description

Zurich is committed to sustainable mobility, investing in high-capacity cycling corridors to reduce car dependence and promote healthier, low-emission travel. The Veloschnellroute Limmattal is a prime example of this initiative, designed to support longer-distance bicycle commutes, particularly with the growth of e-bikes.

However, the future demand for such infrastructure is uncertain, influenced by evolving commuting patterns and urban development. City officials face a critical decision: should they fully commit to an upfront build-out (infrastructure modification) or adopt a flexible, staged modification plan (infrastructure modification with adaptive plan) where future infrastructure modifications depend on usage?

### 2.1.2 Your Work

Your project requires you to develop and justify a proposal for how the infrastructure should be modified now and how it may need to be modified in the future. Your recommendation should consider expected outcomes, benefits and costs, uncertainty about future developments, and your broader vision for how the transport system should function.

To effectively begin this project, start by reviewing the referenced document, "Kosten-Nutzen Analyse Veloschnellroute Limmattal"<sup>3</sup> and the preliminary appraisal conducted last year, i.e., in the pre-study. The first document contains essential background information, useful data, and core assumptions upon which the first report made. The second document can be understood as the result of a preliminary appraisal that you are expected to critically assess, improve and extend. Even though the project is inspired by the reference document(s), you have the freedom to modify the design, the assumptions or the scenarios.

Relevant outcomes, benefits and costs may include, but are not limited to:

- changes in cycling demand and modal share/modal split (e.g. more cycling);
- safety and health benefits associated with increased active mobility (e.g. less accidents);
- environmental and emissions-related benefits (e.g. less noise);
- construction and operating costs.

Main uncertainties may include, but are not limited to:

- the effectiveness of the proposed infrastructure on the points mentioned above;
- spatial development patterns;
- changes in travel demand and commuting behaviour.

An important part of your system model will be how you forecast travel demand and estimate the modal split. You will do this using a classic four-step transport model, an Origin-Destination (OD) Matrix and a utility function. You will need to calculate the share of each mode for each OD pair, for each year, and for each iteration. The modal split will be affected by the timing of the future infrastructure modifications, and the timing of the future infrastructure modifications may be affected by the modal split.

### 2.1.3 Relevant References

- Kanton Zürich, Amt für Verkehr (25.09.2017) Kosten-Nutzen Analyse Veloschnellroute Limmattal (technischer Bericht)

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<sup>3</sup> [Technischer Bericht, KNA Velobahn Limmattal](#)

## 2.2 Specifics for the Forch Mobility Hub Project

### 2.2.1 Description

Zurich is committed to sustainable, multimodal mobility—prioritising strong public-transport nodes that make walking and cycling the favorable access modes to rail and bus (ARE, 2021). Bahnhof Forch (Küsnacht/Maur) is planned to evolve from a simple arrival point into a Mobility Hub (Verkehrsdrehscheibe) with a high-quality hub square, safer and more direct walking/cycling links, secure bike parking (Bike&Ride), sharing offers, and better bus–rail interchange (including the new VBZ 702 connection). A relocation of surface parking into a shared underground garage is envisaged to free public space for active modes and interchange functions.

However, the future demand for such infrastructure is uncertain, influenced by evolving commuting patterns and urban development. City officials now have to decide which infrastructure modification they should implement now and which ones they implement in the future and under what conditions.

### 2.2.2 Your Work

Start by reviewing the official Forch hub document “AP5 Massnahmenblatt GV6 – Küsnacht: Verkehrsdrehscheibe Bahnhof Forch”<sup>4</sup>. It contains essential background information and useful data, that will be helpful when building your model. In addition, consult the reports done by students in previous years, which can be treated as a preliminary appraisal that you are expected to critically assess, improve, and extend using the methods introduced in this year's course. Even though the project is inspired by the reference document(s), you have the freedom to modify the design, the assumptions or the scenarios.

Relevant outcomes, benefits and costs may include, but are not limited to:

- changes in public transport demand and modal share/modal split (e.g. more PT);
- accessibility and travel-time effects (e.g. through better interchange);
- environmental and emissions-related benefits (e.g. less emissions);
- land use effects (e.g. Mobility Hub as commercial hub);
- construction and operating costs.

Main uncertainties may include, but are not limited to:

- the effectiveness of the proposed infrastructure on the points mentioned above;
- spatial development patterns;
- changes in travel demand and commuting behaviour.

An important part of your system model will be how you forecast travel demand and estimate the modal split. You will do this using a classic four-step transport model, an Origin-Destination (OD) Matrix and a utility function. You will need to calculate the share of each mode for each OD pair, for each year, and for each iteration. The modal split will be affected by the timing of the future infrastructure modifications, and the timing of the future infrastructure modifications may be affected by the modal split.

### 2.2.3 Relevant References

- Kanton Zürich (2025). Agglomerationsprogramm Stadt Zürich–Glattal, 5. Generation: Teil 2 – Massnahmenband. Massnahme GV6: Küsnacht – Verkehrsdrehscheibe Bahnhof Forch. Zürich, Switzerland.
- Bundesamt für Raumentwicklung - ARE (2021). Verkehrsdrehscheiben: Gute Beispiele aus der Schweiz und dem Ausland. Bern, Switzerland: ARE.

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<sup>4</sup> [AP Stadt Zürich-Glattal, 5. Generation](#)

- Planwerkstadt AG (2025). Leitbild Fokus Forch: Synthesebericht (Fokus Forch 2045). Zürich, Switzerland: Planwerkstadt AG. Commissioned by Gemeinde Küsnacht, Gemeinde Maur, & Forchbahn AG.

### 3 Project Tasks

The tasks for the project are summarised Table 1, Table 2, Table 3, and Table 4. More detailed descriptions of the tasks will be provided for each Phase of the project in their dedicated project sheet.

Table 1. Tasks

| Week |                                                                                                                                                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Connection to SIA (not all SIA parts are done)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No.  | Main focus                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1-2  | <p>Build your team</p> <p>Introduce the situation and make your case for change</p> <p>Explain how you would appraise the infrastructure modifications and adaptive plans</p> | <p>Build your team. Select your project from 1) Cycling infrastructure, 2) Mobility Hubs.</p> <p>Describe the trends and initiatives at Europe, Switzerland and Canton levels that will help you argue why your system modification is required. (You can use LLMs to help you identify the sources and the material within these sources that are useful to you).</p> <p>Describe your problem at the meso, macro, and micro levels, e.g., what is the canton concretely trying to achieve, what is the municipality trying to achieve, and what are you specifically trying to achieve with your project.</p> <p>Identify the stakeholders who will have an opinion on whether your proposed infrastructure modification is approved.</p> <p>Explain the types of outcomes, benefits and costs you expect, e.g., increased numbers of cyclists, the improvement in health, the amount of money required.</p> <p>Explain the criteria you will use to determine the infrastructure modification and adaptive plan you will suggest.</p> | <p><i>SIA102-110</i></p> <ul style="list-style-type: none"> <li>- <i>Conducting market assessment and analysis</i></li> <li>- <i>Identifying the potential effects and interactions on the designed living space in relation to society, nature, technology, and the economy</i></li> <li>- <i>Completing the list of stakeholders, documenting their needs</i></li> <li>- <i>Identifying relevant environmental aspects in accordance with legislation</i></li> </ul> <p><i>SIA102-112</i></p> <ul style="list-style-type: none"> <li>- <i>Clarifying the overarching goals and constraints</i></li> <li>- <i>Identifying needs, “functional, economic and design” goals, constraints, and requirements “and sustainability priorities”</i></li> <li>- <i>Proposing and deciding on the evaluation procedure and evaluation criteria</i></li> </ul> |

Table 2. Tasks

| Week |                                                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Connection to SIA (not all SIA parts are done)                                                                                                                                                                                                                  |
|------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No.  | Main focus                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                 |
| 3-4  | Model the basic situation                                                                 | <p>Explain the infrastructure modifications and adaptive plans being considered. These will include the ones investigated in the pre-study. They can include others if you wish.</p> <p>Develop a model of the system including the interactions of the system and the environment, and how it evolves over time with particular attention to the outcomes to be used to evaluate the possible system modifications as well as the costs and benefits related to these modifications.</p> <p>Use Python.</p>                                                           | <p><i>SLA102 – 112</i></p> <ul style="list-style-type: none"> <li>- <i>Identifying and evaluating the effects of the possible system modifications, including opportunities and risks</i></li> </ul>                                                            |
| 5-6  | Expand the model to consider uncertainty                                                  | Expand the model of the system to include models of the uncertain variables and how they change over time. For example, the distributions used for the uncertain variables might be normal distributions.                                                                                                                                                                                                                                                                                                                                                              | <p><i>SLA102 – 112</i></p> <ul style="list-style-type: none"> <li>- <i>Identifying and evaluating the effects of the possible system modifications, including opportunities and risks</i></li> </ul>                                                            |
| 7-8  | Expand the model to evaluate the possible infrastructure modifications and adaptive plans | Explain the model of the system to include the proposed infrastructure modifications (and associated future modifications if appropriate) and how they are envisioned to meet future needs. Include when the modifications are to happen and if the infrastructure following the modifications is considered robust or flexible or something in between. Also include the indicators to be used to trigger the modifications, as well as the triggering values, and perhaps a discussion of lag time between triggering a modification and when it is likely to occur. | <p><i>SLA102 – 112</i></p> <ul style="list-style-type: none"> <li>- <i>Identifying and evaluating the effects of the possible system modifications, including opportunities and risks</i></li> <li>- <i>Presenting possible system modifications</i></li> </ul> |

Table 3. Tasks

| Week |                                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Connection to SIA (not all SIA parts are done)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No.  | Main focus                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 9-10 | Appraise the possible infrastructure modifications and adaptive plans | <p>Appraise the possible infrastructure modifications using outcomes and costs and benefits.</p> <p>When focusing on costs and benefits</p> <ul style="list-style-type: none"> <li>- Assign a monetary value to each performance indicator in your objective function (e.g., travel time savings, reduced emissions). These values are to be justified through a literature review or your own estimations.</li> <li>- Include a sensitivity analysis for the range of values you put on each of the outcome units</li> <li>- Include additional outcomes that are difficult to monetise as part of the argumentation.</li> </ul> <p>Your analysis and argumentation should also include:</p> <ul style="list-style-type: none"> <li>- Identification of plans that perform well over multiple scenarios (i.e., robustness).</li> <li>- Identification of the key uncertainties in the analysis and scenarios that provide specific risks and opportunities.</li> <li>- Identification of the most relevant monitoring indicators and best thresholds for the adaptive plans.</li> </ul> | <p><i>SIA102 – 112</i></p> <ul style="list-style-type: none"> <li>- <i>Identifying and evaluating the effects of the possible system modifications, including opportunities and risks</i></li> </ul> <p><i>SIA102 – 113</i></p> <ul style="list-style-type: none"> <li>- <i>Verifying the plausibility of the profitability analysis, considering opportunities and risks and the entire life cycle</i></li> <li>- <i>Estimating the financial requirements of the possible system modifications based on empirical values</i></li> <li>- <i>Highlighting the impact of changes in specifications</i></li> <li>- <i>Estimating the life cycle costs</i></li> <li>- <i>Assessing the compatibility of costs and economic efficiency with the specifications (costs, profitability, key figures)</i></li> </ul> |

Table 4. Tasks

| Week  |                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Connection to SIA (not all SIA parts are done)                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No.   | Main focus                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 11-12 | Proposal                  | <p>Make a proposal for the infrastructure modifications and adaptive plan that you think should be selected and why, using costs and benefits and outcomes.</p> <p>Compare the different infrastructure modifications and adaptive plans by analysing the expected values from your Monte Carlo simulations, cumulative distribution curves, or disaggregated costs</p> <p>For the system modifications propose a schedule, keeping in mind both time and the evolution of system condition over time</p> | <p><i>SLA102 – 114</i></p> <ul style="list-style-type: none"> <li>- <i>Setting the schedule requirements</i></li> <li>- <i>Verifying the plausibility of the time required, taking into account opportunities and risks</i></li> </ul> <p><i>SLA102 - 115</i></p> <ul style="list-style-type: none"> <li>- <i>Developing the basis for decisions on how to proceed</i></li> <li>- <i>Defining the system modification to be implemented</i></li> </ul> |
| 13    | Three slide presentations | Prepare and present three slides illustrating a specific improvement you made to the modelling and how it affected your opinion.                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 14    | Outlook                   | Discussion on the course, presentations from doctoral researchers on advanced modelling                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

#### 4 How to use the Project Sheets

This project sheet covers Weeks 1 and 2. Subsequent phases (Weeks 3–11) will have dedicated sheets on Moodle and GitHub. Each sheet pairs conceptual report tasks with practical Jupyter Notebooks (.ipynb), divided into core requirements and optional extensions.

We distinguish between:

##### **Tutorials/Supporting Tasks (optional)**

Designed to help you understand the modelling framework, become familiar with the Jupyter notebooks, and develop an intuition for the rationale behind each analysis. These tasks do not directly produce outputs for the final report.

However, especially for students with limited experience in Python or quantitative modelling, we strongly recommend completing them.

##### **Mandatory/Expected Task**

Specifying core activities and deliverables that every group is expected to complete and include in the final report.

##### **Advanced Task (optional)**

Suggesting extensions for groups interested in exploring additional analyses, testing alternative assumptions or methods, or gaining a deeper understanding of the framework.



The exercises will generally follow this structure: first letting you explore and understand the modelling framework, then asking you to perform mandatory analyses, and finally giving you hints to further extend your approach.

## 5 Tasks for Week 1 and 2

### 5.1 Team & Project Selection

#### Mandatory Task

List your team members and confirm your project choice (Limmattal Cycling infrastructure or Forch Mobility Hub). Further information about the two alternatives is presented below.

Once you've decided which project to work, please send the full list of team members and selected topic to Orlando (roman@ibi.baug.ethz.ch) before the start of the next lecture (22.09.26).

### 5.2 Policy context and problem definition

#### Mandatory Task

To contextualise your analysis, identify relevant policies, strategies, and planning initiatives at the governance levels that support the reasoning for the proposed infrastructure intervention. We do not expect you to provide an exhaustive overview of the Swiss policy landscape. Instead, select the most relevant policies and planning initiatives that help frame the problem, justify the proposed intervention, and define policy-relevant objectives for your specific case study.

- European level: e.g. EU Cycling Declaration, Sustainable and Smart Mobility Strategy.
- National level: e.g. Velogesetz, Perspektive Bahn 2050, Transport Outlook 2050.
- Cantonal level: e.g. Kantonaler Richtplan, cantonal transport, cycling, or climate strategies.
- Regional level: e.g. regional mobility plans, Agglomeration programmes, corridor strategies.
- Local level: e.g. municipal transport plans, land use plans.

Use this information to describe the problem at (at least) three nested scales, which are not necessarily equivalent to the spatial scales presented above:

- Macro: what broader (national, European and global ) transport and climate challenge does this corridor sit within?
- Meso: what specific commitment has the canton, region or municipalities made that this project helps deliver?
- Micro: what are you trying to achieve with your project?

### 5.3 Define expected outcomes, benefits and costs

#### Mandatory Task

Begin by defining the expected outcomes of your proposed infrastructure intervention. Think about the different dimensions that may be affected by your project and identify the changes you expect to observe over the planning horizon, when compared to the current state. At this stage, you are not expected to quantify these changes, but rather to formulate hypotheses about the direction of change and identify the aspects that are most relevant to evaluate.

Consider outcomes across multiple dimensions, such as:

- transport system performance (e.g. modal split, travel demand, travel times, accessibility);
- environmental impacts (e.g. greenhouse gas emissions, greenery/land use, noise);
- economic impacts (e.g. investment costs, operating costs, travel cost savings);
- societal impacts (e.g. health benefits, safety, equity, quality of life).

For each expected outcome, briefly describe:

- the expected direction of change (increase, decrease, or no significant change);
- why it is relevant for your case study;
- whether it is likely to represent a benefit, cost, or potential trade-off.

## 5.4 Evaluation criteria

### **Mandatory Task**

Based on these expected outcomes, define the evaluation criteria that will later be used to compare alternative infrastructure modifications and adaptive planning strategies. For each criterion, specify:

- the indicator(s) that will be used;
- how the criterion will be measured (units, data source, or model output);
- whether it can be directly quantified by the modelling framework or requires external assumptions or qualitative assessment.

This set of evaluation criteria will form the basis for the appraisal of your proposed intervention throughout the remainder of the course.

However, you can always come back to these and further specify them or change them, based on the evolution of your project.

## 5.5 Stakeholder Mapping

### **Mandatory Task**

Identify the stakeholders who may influence, implement, finance, regulate, use, benefit from, or be adversely affected by the proposed infrastructure intervention, with reference to the previously defined objectives and expected outcomes.

Focus on at least five stakeholders representing different categories, such as public authorities, funders, operators, users, businesses, landowners, interest groups, and affected communities.

For each selected stakeholder/group, consider:

- their main interests and objectives;
- the extent and source of their influence;
- their likely level of support, opposition, or uncertainty;
- the type and magnitude of impacts they may experience;
- their role during planning, implementation, operation, and future adaptation;
- potential conflicts, dependencies, or alliances with other stakeholders.

This information will help you in the (qualitative) evaluation of different interventions, which we will focus on in Weeks 9-11.

### **Advanced Task**

Extend the stakeholder analysis using established stakeholder mapping and classification methods. Consider representing stakeholders through influence–interest matrices, salience analyses, and/or network diagrams to better understand relationships, dependencies, conflicts, and collaboration opportunities. Discuss how these insights could inform project governance and how these might link to the defined objectives.

## 6 Resources to get started with GitHub, IDEs and transport modelling

Since students in this course will have different experiences in coding, we provide a set of resources to help you get started. All computational work in the project will be carried out in Python, and we recommend using Visual Studio Code (VS Code) as your development environment and GitHub for version control and collaboration within your group. The course material will be made available through a dedicated GitHub repository.

Students may also have different levels of prior knowledge in transportation modelling. While other courses offered in parallel provide a more detailed introduction to transport theory and its applications, this course does not require you to become a transport modelling expert. We will provide internal material explaining the main components and assumptions of the transport model used for the simulations. For the purposes of the project, you may largely treat the model as a (black-box) simulation engine running in the background: you should understand what goes into the model, what it computes, and how to interpret its outputs, but you are not expected to understand or modify the details of its implementation.

You are encouraged to submit your questions to the Help Centre (Forum) on Moodle where we will try to reply as soon as possible. One-on-one help sessions are also possible on request.

### 6.1 GitHub, VS Code and version control

You will use GitHub to access the project repository, collaborate within your group, keep track of changes, and maintain a reproducible record of your work. If you have not previously worked with Git or GitHub, the following official resources provide a useful introduction:

- **GitHub Docs – Get started with GitHub:** introduction to repositories, Git, commits, branches, and collaboration. [GitHub: Get started with GitHub](#)
- **VS Code – Source Control Quickstart:** practical introduction to cloning a repository, reviewing changes, committing, pulling, and pushing directly from VS Code. [VS Code: Source Control Quickstart](#)

### 6.2 Python and using AI for coding

Much of the exercise material will be provided through Python scripts and Jupyter notebooks. If notebooks are new to you, the official Jupyter introduction provides a simple way to become familiar with their basic structure and execution model. [Project Jupyter: Try Jupyter](#)

Verifying your setup before you start: Once you have cloned the repository and installed the requirements (pip install -r requirements.txt), open your VS Code terminal in the repository root and run:

`python verify_setup.py` (python3 for mac users)

This script performs an automated health check on your environment: it verifies your Python version, tests that all core libraries (including GeoPandas and EMA Workbench) can be loaded properly, and checks that all required model and network datasets are present. Once you see the green success message, you are ready to proceed to Phase 2.

You are encouraged to use AI tools to assist you with coding during the project. They can be particularly useful for understanding unfamiliar parts of the repository, explaining errors, implementing small modifications, plotting or analysis code, and debugging.

VS Code provides integrated support for AI-assisted coding, and its official documentation provides a useful starting point: [VS Code: AI-assisted development](#) (you might want to consider that [ETH provides free access to Google's Gemini or Microsoft Copilot](#)).

AI should, however, be used as an assistant rather than as a substitute for understanding and verification. A useful workflow is:

understand the task → ask AI for assistance (especially to understand the relationships between different files) → inspect the proposed changes → run and test the code → check whether the outputs make sense → commit the validated changes → push to the shared GitHub repository

Do not assume that code is correct simply because it runs. You remain responsible for the correctness of your implementation and analysis, and the quality and correctness of the submitted code form part of the final assessment.

### 6.3 Understanding the transport model

We will provide dedicated course material explaining the transport model used in the project at the beginning of Phase 2 (week 3 of the semester), including its main inputs, assumptions, mechanisms, and outputs. The MehrSpur worked example and the technical walkthroughs included in the exercise sheets will also show how the model is used within a broader infrastructure-planning analysis.

You do not need to understand every equation or implementation detail before starting the project. You should, however, progressively develop enough understanding to answer questions such as:

- What are the main inputs and assumptions of the model?
- Which aspects of the transport system change when an infrastructure modification or policy is introduced?
- Which outputs are used to evaluate performance?
- What important real world processes are simplified or omitted?
- How should these limitations affect the interpretation of the results?

This level of understanding is particularly important when interpreting simulation results and when deciding whether additional quantitative or qualitative analysis is required.

### 6.4 A note on computation time

Some of the analyses performed during the semester will require running the simulation model repeatedly across plans, scenarios and multiple years. Depending on the analysis, a computational experiment may take from a few minutes to several hours.

You should therefore plan computationally intensive analyses in advance rather than leaving them until immediately before a helper session, presentation, or submission deadline. Whenever possible, first test your code using a small number of simulations, verify that the workflow and outputs are correct, and only then launch the complete experiment.